NAME ☞	
SCHOOL ☞	
TEACHER ☞	

Pre-Junior Certificate Examination, 2019

Mathematics

Paper 2

Ordinary Level

Time: 2 hours

300 marks

School Stamp

Running Total	

For Examiner			
Question	Mark	Question	Mark
1		11	
2			
3			
4			
5			
6			
7			
8			
9			
10		Total	

Grade

Instructions

There are 11 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

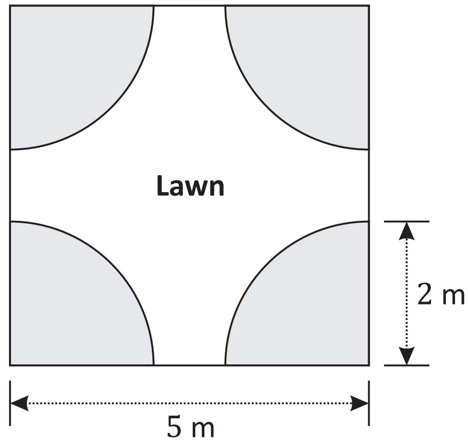
You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

Question 1 **(Suggested maximum time: 10 minutes)**

Tom's garden is in the shape of a square with sides of length 5 m. There are flower beds at each of the four corners of the garden. Each of the flower beds is in the shape of a sector, of radius 2 m. The remaining area of the garden makes up the lawn, as shown.



(a) Find the **area** of the garden in m^2.

(b) Find the **total area** of the 4 flower beds.
Give your answer in m^2, correct to one decimal place.

(c) Find the **area** of the lawn.
Give your answer in m^2, correct to one decimal place.

(d) Tom estimates that the **perimeter** of his lawn is 17 m.
Show that he is correct.

Question 2 **(Suggested maximum time: 15 minutes)**

(a) The Spire, also called the Monument of Light, is a large monument in Dublin city centre. It was completed in January 2003. It is 120 m tall and 3 m wide at the base.

(i) How many years is it since the Spire was completed?

The Spire is made up of 8 sections, sitting directly on top of each other. Ruth says that the height of each section is 20 m.

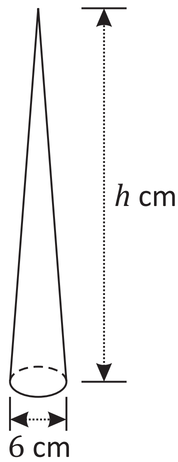
(ii) Is Ruth correct? Justify your answer.

Answer:

Justification:

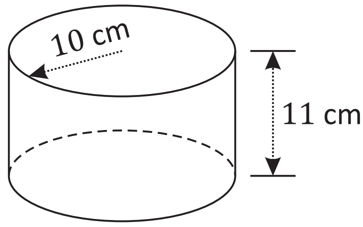
A class makes a scale model of the Spire for a project. The model is 6 cm wide at the base.

(iii) Find h, the height of the scale model in cm.

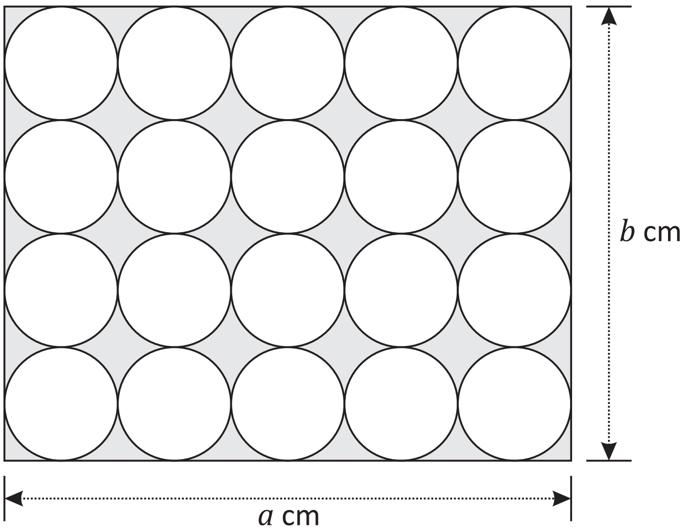


(b) A cylinder has a radius of 10 cm and a height of 11 cm.

(i) Find the **volume** of the cylinder. Give your answer, in cm^3, correct to one decimal place.



Twenty of the cylinders are packed in a rectangular box of height 11 cm.
The diagram shows how the cylinders are arranged inside the box of length a cm and width b cm. The shaded area is the space between the cylinders.



(ii) Work out the values of a and b.

$a =$ ______ $b =$ ______

(iii) Find the **volume** of the box, in cm^3.

(iv) Work out the volume of the shaded area inside the box that is <u>not</u> filled by the cylinders.
Give your answer correct to two significant figures.

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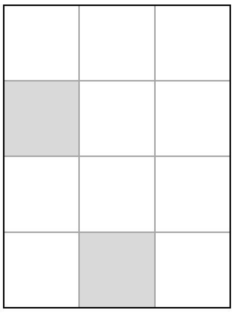
Question 3 **(Suggested maximum time: 10 minutes)**

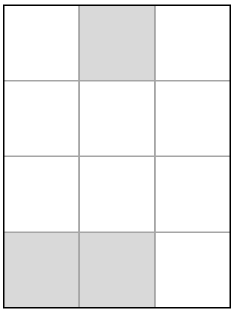
In a game, a counter is thrown at random onto one of four boards.

The four different boards used in the game are labelled **A**, **B**, **C** and **D** as shown below.

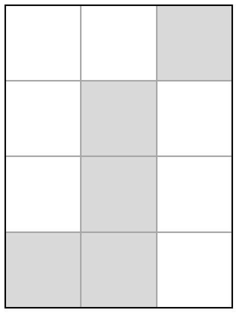
The counter will always land exactly on one square.
If you land on a grey square you win a prize. There are 12 squares on each board.



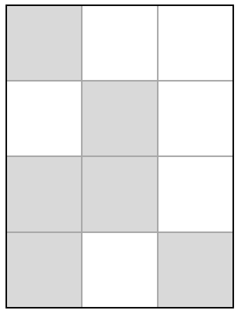
Board A



Board B



Board C



Board D

(a) Fill in the table to show the probability of winning a prize on each of the boards.

Event	Probability
Winning a prize on **Board A**	
Winning a prize on **Board B**	
Winning a prize on **Board C**	
Winning a prize on **Board D**	

(b) On which board do you have an equal chance of winning or not winning a prize?
Justify your answer.

Answer:

Justification:

(c) Which board would you choose to play on if you wanted the best chance of winning a prize?
Justify your answer.

Answer:

Justification:

(d) Which other board gives you the nearest chance of winning to that of Board **B**?
Justify your answer.

Answer:

Justification:

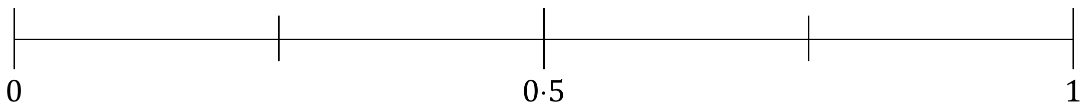
(e) Jake plays the game on **Board A** 24 times.
How many times would you expect him to win a prize?

Question 4 **(Suggested maximum time: 5 minutes)**

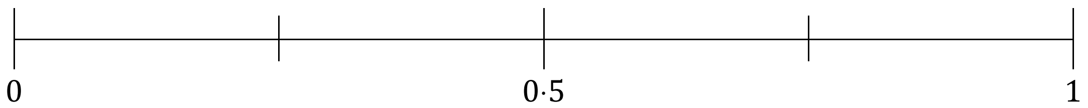
(a) **(i)** Dani says that the probability that she will be late for school tomorrow is $1 \cdot 5$.
Explain why Dani's statement is incorrect.

(ii) Eoin plays a hurling match. The probability that he scores a point in the match is $0 \cdot 7$.
What is the probability that Eoin **does not** score a point in the match?

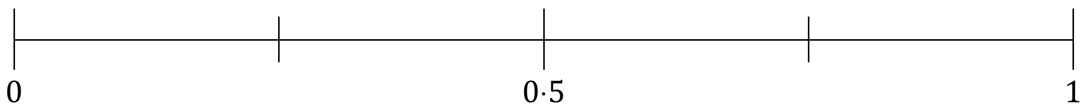
(b) **(i)** **Write** the letter **X** in the correct place on the probability scale below to show the probability that it will rain on at least one day in Ireland next month.



(ii) **Write** the letter **Y** in the correct place on the probability scale below to show the probability that you will get a seven when you roll a fair dice.

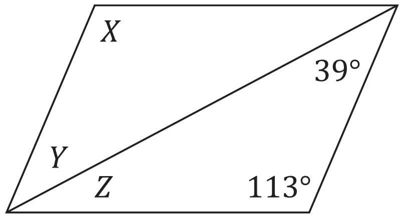


(iii) **Write** the letter **Z** in the correct place on the probability scale below to show the probability that you will get tails when you toss a coin.



Question 5 **(Suggested maximum time: 5 minutes)**

(a) The diagram shows a parallelogram.
The sizes of some of the angles are marked.

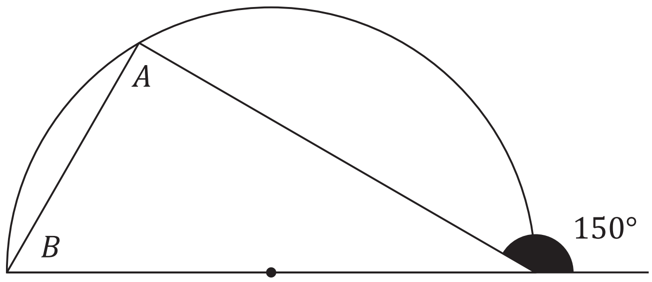


(i) Write down, in your own words, any theorem you know about parallelograms.

(ii) Find the size of the angle X, the angle Y and the angle Z, without measuring.

$|\angle X| =$ $\boxed{}$ $|\angle Y| =$ $\boxed{}$ $|\angle Z| =$ $\boxed{}$

(b) The diagram shows a triangle standing on a semicircle.

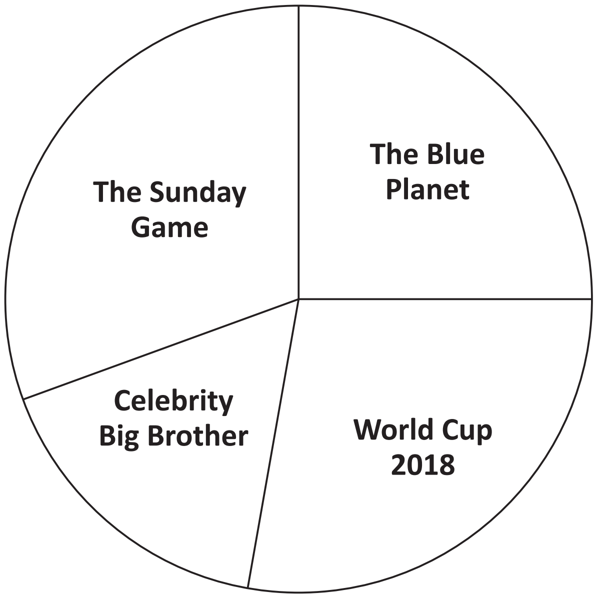


Find the size of the angle A and the angle B, without measuring.

$|\angle A| =$ $\boxed{}$ $|\angle B| =$ $\boxed{}$

Question 6 **(Suggested maximum time: 15 minutes)**

A group of 72 students were asked in a survey to name their favourite programme shown on TV last summer. The results are shown in the pie chart below.



(a) Use a protractor to measure the **size of the angle** in each sector of the pie chart.

Sector	The Blue Planet	The Sunday Game	Celebrity Big Brother	World Cup 2018
Size of angle (°)				

(b) In this survey, students were asked to name their favourite programme on TV.
Put a tick in the correct box to show what type of data this is. **Justify** your answer.

Type of data: categorical ☐ numerical ☐
(tick (✓) **one** box only)

Justification:

(c) What percentage of students named Celebrity Big Brother as their favourite programme?
Give your answer correct to one decimal place.

(d) Megan says that 18 students named The Blue Planet as their favourite programme. Show that she is correct.

(e) Complete the table below to show the number of students who named each programme as their favourite programme.

Programme	The Blue Planet	The Sunday Game	Celebrity Big Brother	World Cup 2018
Number of students	18			

Question 7 **(Suggested maximum time: 10 minutes)**

The speeds, in km/h, of 15 vehicles recorded by a speed camera are shown below.

33	45	55	49	32
43	56	23	56	56
31	60	52	49	27

(a) Complete the stem and leaf diagram below to show this data.

2									
3									
4									
5									
6									

Key: | 3 | 2 | = | |

(b) Work out the **median** speed of this data.

(c) Work out the **mode** of this data.

(d) Work out the **mean** speed of this data.
Give your answer correct to one decimal place.

(e) Work out the **range** of speeds of this data.

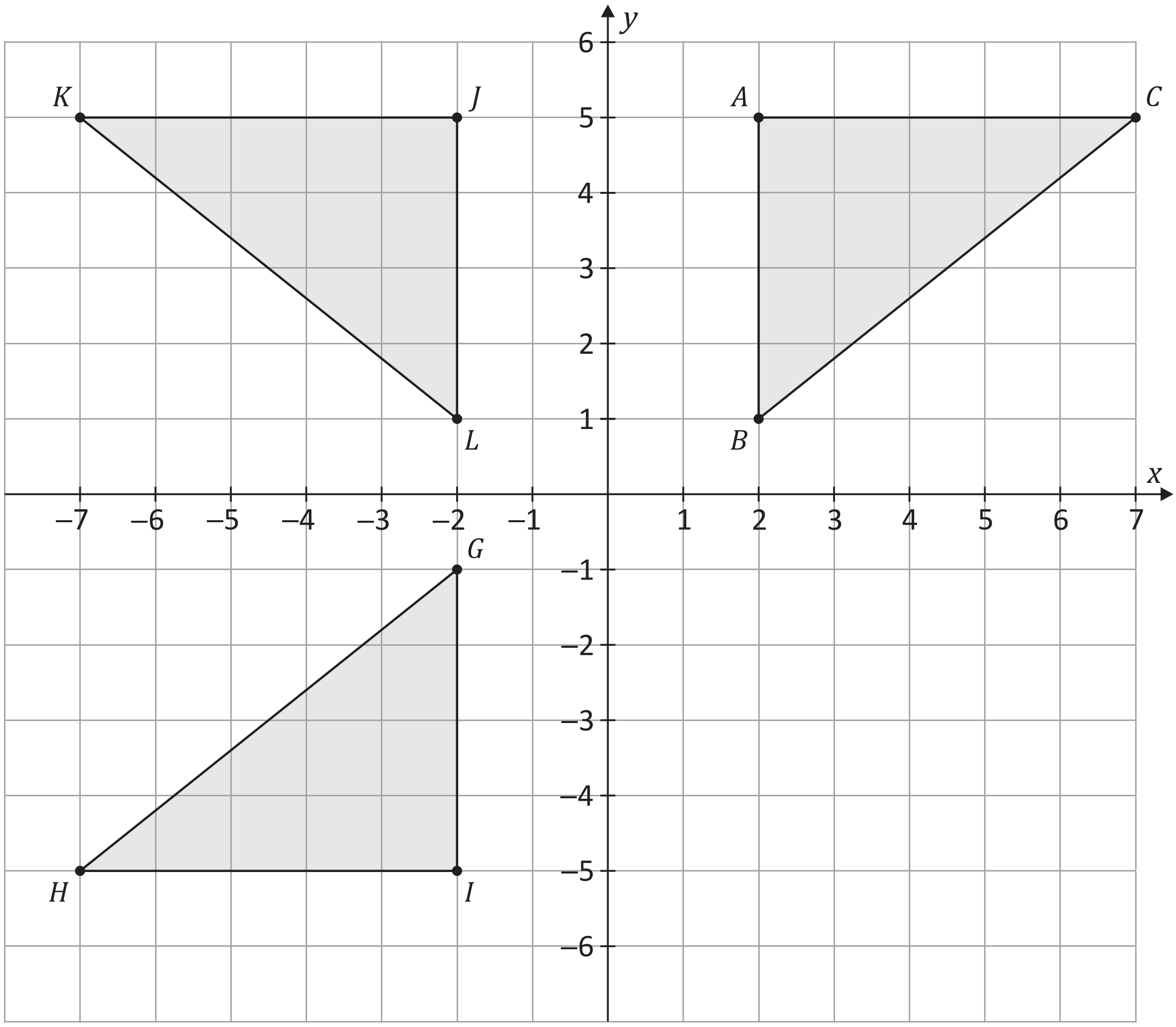
(f) The speed of the next vehicle that the speed camera recorded was 59 km/h.
How will this affect the range? Tick (✓) one box only.
Give a reason for your answer.

range will increase	range will decrease	range will stay the same
☐	☐	☐

Reason:

Question 8 **(Suggested maximum time: 15 minutes)**

Three congruent right-angled triangles are shown on the co-ordinated diagram below.



(a) Write down the co-ordinates of the points A, B and C.

$A =$ (,) $B =$ (,) $C =$ (,)

(b) Using the diagram, or otherwise, find $|AB|$.

$|AB| =$

(c) Using the diagram, or otherwise, find $|AC|$.

$|AC| =$

(d) Use the theorem of **Pythagoras** to find $|BC|$, the **length** of $[BC]$.
Give your answer in the form $\sqrt{n}$ where $n \in \mathbb{N}$.

(e) Write down the co-ordinates of the midpoint of $[BC]$.

(f) On the co-ordinated diagram, draw the triangle DEF, the image of triangle ABC under axial symmetry in the x-axis.

(g) Complete each of the following statements correctly.

(i) Triangle ABC is the image of triangle ______ under axial symmetry in the y-axis.

(ii) Triangle DEF is the image of triangle ______ under central symmetry in (0, 0).

(iii) Triangle ABC is the image of triangle ______ under a rotation.

Question 9 **(Suggested maximum time: 5 minutes)**

(a) The line l has a slope of 2. It goes through the point $(3, -1)$.
Write down the **equation** of the line l, in the form $y = mx + c$.

(b) Is the point $(-1, 2)$ on the line m ($y = -3x + 5$)? **Justify** your answer.

The point $(-1, 2)$: (tick (✓) **one** box only)	**is** on m ☐	is on **not** m ☐

Justification:

(c) Are line l and line m perpendicular to each other? **Justify** your answer.

Line l and line m are: (tick (✓) **one** box only)	perpendicular ☐	**not** perpendicular ☐

Justification:

Question 10 **(Suggested maximum time: 10 minutes)**

The diagram shows a road sign indicating that the road ahead narrows on both sides.

(a) How many **axes of symmetry** does the road sign have?

	1	2	3	4
Axes of symmetry: (tick (✓) **one** box only)	☐	☐	☐	☐

(b) The road sign is in the shape of an equilateral triangle.
Write down the size of each of the angles in the triangle.

(c) **Construct** an equilateral triangle with side of length 7·5 cm.
Show all of your construction lines clearly.

Construction:

Question 11 **(Suggested maximum time: 10 minutes)**

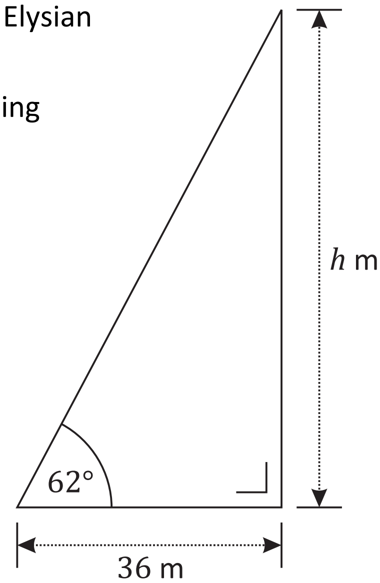
(a) Use a **calculator** to find the value of each of the following.
Give each answer correct to two decimal places.

(i) $\cos 46° =$

(ii) $\sin 32° =$

(b) Kellie wants to find the height of The Elysian building in Cork.

The distance from where she is standing to the base of the building is 36 m.
She measures the angle of elevation of the top of the building to be 62°, as shown.



(i) Write down the value of $\tan 62°$ as a fraction, in terms of h.

$\tan 62° =$

(ii) Use a **calculator** to find the value of $\tan 62°$.
Give your answer correct to one decimal place.

$\tan 62° =$

(iii) Use your answers from part **(i)** and part **(ii)** to find h.
Give your answer correct to the nearest metre.

Page for extra work.
Label any extra work clearly with the question number and part.

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Label any extra work clearly with the question number and part.

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